

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE																																
	Supplementary Winter-2023																																
	Course: B. Tech. Branch :Computer Engineering and Allied					Semester :IV																											
	Subject Code & Name: BTBSC404 Probability &Statistics																																
	Max Marks: 60			Date:23/01/24			Duration: 3 Hr.																										
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.																																
								(Level /CO)	Marks																								
Q. 1	Solve Any Two of the following.									12																							
A)	There are 3 true coins and 1 false coin with ‘head’ on both sides. A coin is chosen at random and tossed 4 times. If ‘head’ occurs all the 4 times, what is the probability that the false coin has been chosen and used?								CO-2	6																							
B)	If A and B are any 2 events such that $P(A) = \frac{3}{4}$ and $P(B) = \frac{5}{8}$. Prove that $\frac{3}{8} \leq P(A \cap B) \leq \frac{5}{8}$								CO-2	6																							
C)	A box contains 5 green pencils and 7 yellow pencils. Two pencils are chosen at random from the box without replacement. What is the probability that both are yellow?								CO-2	6																							
Q.2	Solve Any Two of the following.									12																							
A)	Find the value of k , if the following function is a probability density function. $f(x) = \begin{cases} k(x - 1)^3 & 1 \leq x \leq 3 \\ 0 & otherwise \end{cases}$								CO-1	6																							
B)	A firm has two cars which it hires out day by day. The number of demands of a car each day is distributed as a Poisson variate of mean 1.5. Calculate the probable number of days in a year on which (i) neither car is in demand (ii) a demand is refused.								CO-1	6																							
C)	Five coins are tossed 100 times and the following results were obtained <table border="1"><tr><td>No. of heads</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>frequency</td><td>10</td><td>20</td><td>30</td><td>15</td><td>15</td><td>10</td></tr></table> Fit a Binomial distribution.								No. of heads	0	1	2	3	4	5	frequency	10	20	30	15	15	10	CO-1	6									
No. of heads	0	1	2	3	4	5																											
frequency	10	20	30	15	15	10																											
Q. 3	Solve Any Two of the following.									12																							
A)	Calculate Karl Pearson’s co-efficients of correlation from the following data <table border="1"><tr><td>x</td><td>28</td><td>45</td><td>40</td><td>38</td><td>35</td><td>33</td><td>40</td><td>32</td><td>36</td><td>33</td></tr><tr><td>y</td><td>23</td><td>34</td><td>33</td><td>34</td><td>30</td><td>26</td><td>28</td><td>31</td><td>36</td><td>35</td></tr></table>								x	28	45	40	38	35	33	40	32	36	33	y	23	34	33	34	30	26	28	31	36	35	CO-3	6	
x	28	45	40	38	35	33	40	32	36	33																							
y	23	34	33	34	30	26	28	31	36	35																							

B)	Calculate the value of rank correlation coefficient from the following data regarding marks of six students in statistics and accountancy in a test <table><tr><td>Marks in statistics</td><td>40</td><td>42</td><td>45</td><td>35</td><td>36</td><td>39</td></tr><tr><td>Marks in accountancy</td><td>46</td><td>43</td><td>44</td><td>39</td><td>40</td><td>43</td></tr></table>	Marks in statistics	40	42	45	35	36	39	Marks in accountancy	46	43	44	39	40	43	CO-4	6				
Marks in statistics	40	42	45	35	36	39															
Marks in accountancy	46	43	44	39	40	43															
C)	Prove that limits of correlation coefficients are lies between $-1 \leq r \leq 1$.	CO-4	6																		
Q.4	Solve Any Two of the following.		12																		
A)	Find the equations of the lines of regression on (i) Y on X and (ii) Xon Y and also a co-efficient of correlation from the following table. <table><tr><td>X</td><td>62</td><td>64</td><td>65</td><td>69</td><td>70</td><td>71</td><td>72</td><td>74</td></tr><tr><td>Y</td><td>126</td><td>125</td><td>139</td><td>145</td><td>165</td><td>152</td><td>180</td><td>208</td></tr></table>	X	62	64	65	69	70	71	72	74	Y	126	125	139	145	165	152	180	208	CO-3	6
X	62	64	65	69	70	71	72	74													
Y	126	125	139	145	165	152	180	208													
B)	The equations to the two lines of regression are $6y = 5x + 90$ and $15x = 8y + 130$. Find (i) the means of x and y , (ii) the coefficient of correlation ,(iii) if variance of $x = 16$, find also the standard deviation of y .	CO-3	6																		
C)	If the coefficient of correlation between two variables x and y is 0.5 and the acute angle between their lines of regression is $\tan^{-1}\left(\frac{3}{5}\right)$. Prove that $\sigma_x = \frac{1}{2}\sigma_y$.	CO-3	6																		
Q. 5	Solve Any Two of the following.		12																		
A)	A manufacturer claims that only 4% of his products supplied by him are defective. A random sample of 600 products contained 36 defectives. Test the claim of the manufacturer.	CO-4	6																		
B)	A machine produced 16 defectives articles in a batch of 500. After overhauling it produced 3 defectives in a batch of 100. Has the machine improved?	CO-4	6																		
C)	A soap manufacturing company was distributing a particular brand of soap through a large number of retail soap. Before a heavy advertisement campaign, the mean sale per week per shop was 140 dozens. After the campaign a sample of 26 shops was taken and the mean sale was found to be 147 dozens with standard deviation of 16. Can you consider the advertisement effective?	CO-5	6																		
	*** End ***																				